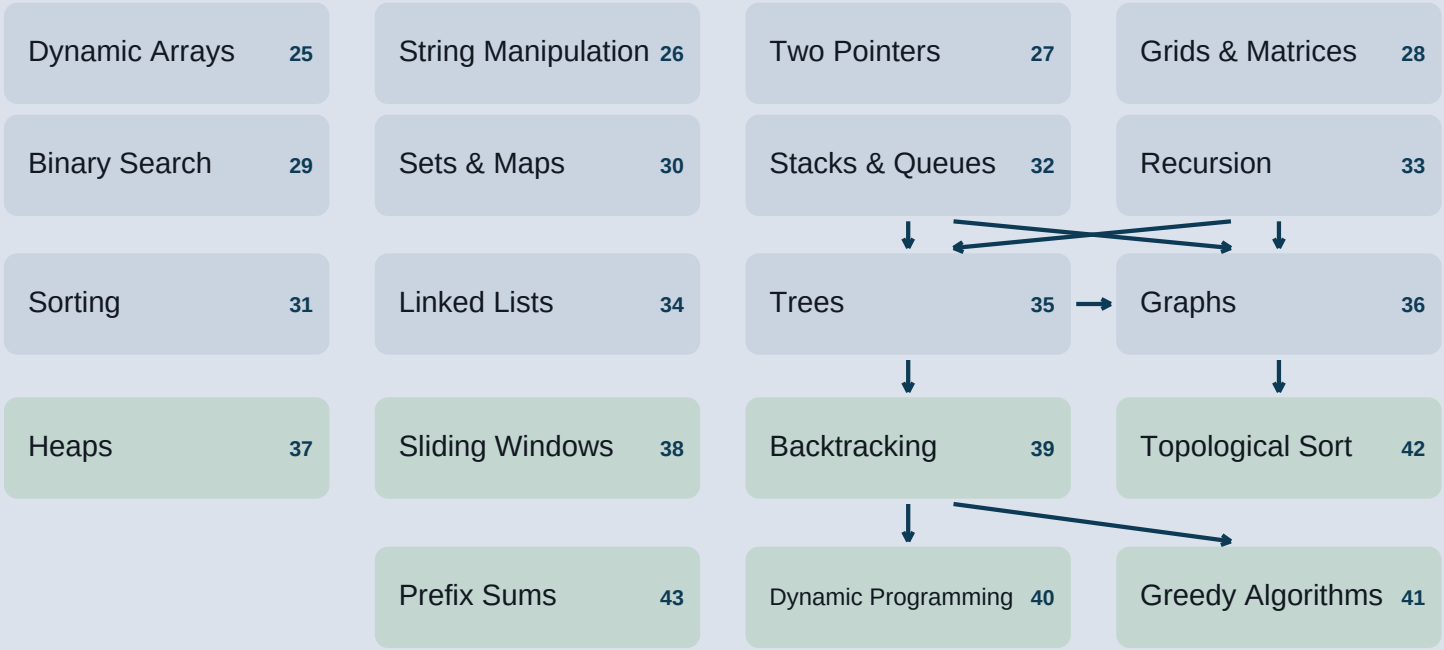
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Tier 1 - Fundamental
Tier 2 - Common and important

Interview checklist

1 Understand the problem

- Read the statement twice
- Examples out loud
- Clarify edge cases, I/O and scale

2 Design the algorithm

- Sketch the naive solution
- Upper & lower bounds
- Look for triggers
- Use boosters

3 Explain the solution

- Examples
- Indented English
- Name & justify

4 Get buy-in

Should I start coding this up,
or would you like me to keep
looking for something better?

5 Code the solution

- Main logic
- Helper functions
- Stop if you get lost

6 Verify the solution

- Top-to-bottom pass
- Tricky expressions
- Trace a tiny input
- Check edge cases
- Verify time & space

Boosters - Help Developers Crack Algorithms

Lower: unavoidable work/output. Upper: correct algorithm cost; start naive.
Constraints: feasible time budget; TLE = time limit exceeded.

B - Optimise brute force: find the bottleneck

-> Preprocess: compute once, reuse
-> Better data structure: speed up the costly operation
-> Skip work: compute only what's needed

H - Hunt for properties
D - Decrease the difficulty
C - Cycle through the catalog
A - Articulate your blocker

Big O of ...

Complexity / typical growth	Examples	Approx. max n conservative / generous
O(1) - Constant (one) Bounded work as n grows.	Array access min(n, 10)	No n-based limit
O(log n) - Logarithmic (log n) Double n: add constant work.	Binary search log2(n), log3(n)	Very large
O(n) - Linear (n) Double n: about 2x work.	Full scan; 3*n + 5 Practice cue: n = 200,000	2 million / 200 million
O(n log n) - Linearithmic (n log n) Double n: slightly more than 2x work.	Merge sort n*log2(n) + n	~100,000 / ~8 million
O(n^2) - Quadratic (n squared) Double n: about 4x work.	All pairs n*(n-1)/2, 1+2+...+n	~1,000 / ~14,000
O(n^3) - Cubic (n cubed) Double n: about 8x work.	Three full nested loops n^3 + n^2*log2(n)	~100 / ~500
O(2^n) - Exponential (two to the n) Add 1 to n: about 2x work.	All subsets (count) 2^n + n^2	20 / 27
O(n!) - Factorial (n factorial) Add 1 to n: multiply work by n+1.	All permutations (count) 1*2*...*n	9 / 11

Rounded size cues, not guarantees. Start with conservative estimates.